

Teacher Guide — senior / module-02- panchabhuta

IKS Curriculum

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Overview

Module 2 — Pañcabhūta: Sāṅkhya Cosmology and Cross-Cultural Comparison

Age band: Senior · **Grades:** 9–12 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Recall** the *pañca-mahābhūta* and locate them within the 25-*tattva* enumeration of Sāṅkhya philosophy.
2. **Quote** at least one primary-source verse (in IAST with translation) on the cosmological emergence of the elements.
3. **Compare and contrast** the Sāṅkhya five-element scheme with (a) Empedocles' four elements, (b) Aristotle's *quintessence*, and (c) the modern atomic/molecular framework — articulating what each captures that the others do not.
4. **Analyse** the relationship between *bhūta* and *tanmātra* using the quality-count rule, and explain why the framework is internally consistent independent of modern science.
5. **Construct** an original argument (essay or research paper) on whether classical Indian intellectual frameworks should be taught alongside modern science in school — citing primary sources, secondary scholarship, and at least one counter-argument.

NEP 2020 alignment

- **§4.6** — Integration of IKS into mainstream education.
- **§11.8** — Multidisciplinary linkage (philosophy + science + history).
- **§4.23** — Inquiry-based and experiential learning.
- **§22** — Indian languages, arts, and culture in the school curriculum.

Prerequisites

- Familiarity with Middle-band Module 2 content (the five elements at conceptual level).

- Comfort reading short Sanskrit verses with provided transliteration and translation.
- Basic familiarity with the periodic table and states of matter.
- Ability to write a structured 1–2 page argumentative essay.

Key vocabulary

See `_shared/glossary.md` for base entries. Senior-band additions:

- *tattva* — “thatness, principle.” The 24 (or 25) principles of Sāṅkhya philosophy. The *bhūtas* are the last five.
- *prakṛti* — “nature, primordial matter.”
- *puruṣa* — “person, consciousness.”
- *guṇa* (Sāṅkhya sense) — *sattva*, *rajas*, *tamas* — the three constituent qualities of *prakṛti*.
- *avyakta* / *vyakta* — unmanifest / manifest.

Materials checklist

- Printed primary-source extracts: SB 2.10.3, 3.5.36, 3.26.36, 3.26.44, 3.26.50 in IAST + translation (from cache)
- Reading: short extract from Larson, *Classical Sāṅkhya* (1979) — teacher-prepared synopsis
- Reading: short extract from Plofker, *Mathematics in India* (2009) — historical context
- Optional reading: extract from Aristotle’s *On Generation and Corruption*

Day-by-day overview

Day	Title	Focus	Assessment
1	Hook: A Persistent Question	What does it mean to “categorise matter”?	Discussion
2	The Sāṅkhya Tattva Ladder	The 25 principles in context	Diagram quiz
3	Quality-Count Rule and the Tanmātras	Internal consistency of the framework	Concept check
4	Empedocles, Aristotle, and the Greek Four	Cross-cultural comparison	Comparison essay (in-class draft)
5	Mid-module Check: Primary Source Reading	Read SB 3.26.36 in IAST	Formative
6	The Modern Lens: Atoms, Fields, States	What modern science added and dropped	Reflection essay
7	Discussion: Should	Multi-perspective	Group oral assessment

Day	Title	Focus	Assessment
	IKS be in Science Class?	debate	
8	Project Session 1: Research Begins	Choose paper topic + outline	Project draft
9	Project Session 2: Argument Building	Develop counter-arguments	Project draft
10	Final Assessment + Paper Submission	Summative quiz + 5-page paper	Summative + project

Citation sources used

- vidya-karana RAG caches (`.rag-cache/`): SB 2.10.3, 3.5.36, 3.26.36, 3.26.44, 3.26.50, and supporting Bhāgavatam canto 4 / 6 / 11 chunks
- Additional sources in `sources.md` including Larson (1979), Plofker (2009), Aristotle.

Project brief (one-line teaser)

A 5-page argumentative paper: “Should classical Indian intellectual frameworks be taught alongside modern science in school?” Citations required from at least 3 distinct source categories. See `assessments/project-brief.md`.

10-Day Lesson Plan

Module 2 (Senior) — 10-Day Lesson Plan

At a glance

Day	Session	File	Output
1	Hook: A Persistent Question	<code>days/day-01-hook.md</code>	Opening discussion
2	The Sāṅkhya Tattva Ladder	<code>days/day-02-tattvas.md</code>	Diagram of 25 tattvas
3	Quality-Count + Tanmātras	<code>days/day-03-tanmatras.md</code>	Concept check
4	Empedocles, Aristotle, and the Greek Four	<code>days/day-04-greek-comparison.md</code>	Comparison essay draft
5	Primary Source Reading	<code>days/day-05-primary-source.md</code>	Formative quiz
6	The Modern Lens	<code>days/day-06-modern.md</code>	Reflection
7	Discussion: IKS in Science	<code>days/day-07-</code>	Oral assessment

Day	Session	File	Output
	Class	discussion.md	
8	Project Session 1	days/day-08-project-session-1.md	Paper outline
9	Project Session 2	days/day-09-project-session-2.md	Paper draft
10	Final Assessment	days/day-10-final-assessment.md	Summative + paper

Pacing notes

- This is a reading-and-writing heavy module. Build in time for *quiet reading* and *quiet drafting*.
- The Day 5 primary-source reading is the most demanding session — students read a Sanskrit verse with transliteration and grapple with the meaning. Schedule it when energy is highest.
- Day 7 debate can run long. Cap at 35 min.
- Paper drafts (Day 9): students should leave class with at least 2 pages drafted.

What students leave with

- Working vocabulary for Sāṅkhya philosophy (tattva, prakṛti, puruṣa, guṇa).
- Ability to read a short Sanskrit verse with translation and discuss it.
- Comparative analytical writing.
- One original argumentative paper.
- The habit of holding multiple frameworks simultaneously without collapsing them into one another.

Teacher Notes

Teacher Notes — Module 2 Panchabhūta (Senior)

Posture

At Senior band, students can hold complexity. Don't oversimplify. The Sāṅkhya enumeration of 25 *tattvas* is genuinely sophisticated — students at this age can grasp it as a careful philosophical achievement, comparable to Aristotle's *Metaphysics* or Spinoza's *Ethics*.

The honest framing: **the Sāṅkhya scheme is not “early science.” It is metaphysics — a theory about the structure of reality, of which the five gross elements are the last and most concrete layer.** Whether that theory is “true” in any modern sense is a separate question; whether it is *internally consistent and historically influential* is not in dispute.

Common misconceptions at this band

1. **“Sāṅkhya is just five elements.”** No — it is 25 tattvas. The five bhūtas are the last layer (#21–25), preceded by the five tanmātras (#16–20), the eleven indriyas (#5–15), the three antaḥkaraṇa (#2–4), and prakṛti/puruṣa at the foundation. The full enumeration matters for understanding the *bhūta-tanmātra* mapping.
2. **“Sāṅkhya is religious / Vedic.”** Sāṅkhya is one of the six *darśanas* (philosophical systems) of classical Hindu thought, but it is the *most* explicitly non-theistic of them (it does not require a creator God; Yoga as a parallel system adds a theistic dimension). Treat it as philosophy with religious adjacencies, not as religion.
3. **“Empedocles and Sāṅkhya are basically the same.”** Superficially yes (four elements roughly map). Crucially different:
 - Empedocles: four elements + two forces (Love, Strife) drive combination/separation.
 - Sāṅkhya: five elements emerge from a single root (prakṛti) via the unfolding of three *guṇas* (sattva, rajas, tamas); each bhūta is paired with a tanmātra (perceptual quality).
4. **“Aristotle’s quintessence is the same as ākāśa.”** Critical difference: Aristotle’s fifth element was the substance of the *celestial spheres* — heavenly, incorruptible. *Ākāśa* in Sāṅkhya is the universal medium of space-with-sound-as-property — terrestrial, not preserved for the heavens.

On reading primary sources

- Day 5 requires students to actually *read* a Sanskrit verse in IAST. Most won’t have done this before. Schedule it deliberately.
- Don’t ask for pronunciation perfection. Ask for *engagement with the meaning*.
- Have transliteration AND word-by-word gloss AND continuous translation available. The point is to show *what reading the source feels like*, not to teach Sanskrit.

Differentiation hints

- **Tier up:** Senior students with prior philosophy interest can read the Sāṅkhya Kārikā ’s first 6 verses (in translation) and present them to the class.
- **Tier down:** Students less ready for primary-source reading can focus on the comparative essay (Days 4–6) and use translated paraphrase rather than IAST.

Sensitivity notes

- The Day 7 discussion (“Should IKS be in science class?”) will surface strong opinions. Some students will see this as an obvious yes (cultural pride). Some will see it as an obvious no (modernist science). The teacher’s role: *make sure both positions are heard and defended on their best terms*.
- Senior students from minority/non-Hindu backgrounds may feel the question is loaded. Acknowledge this; frame as a public-policy question, not a faith question.

Project oversight

- The paper is the major output. 5 pages is *not* a short paper at Senior band — it requires real research time.
- Confirm topics by Day 7. Discourage topics that are too broad (“Indian science is great”) in favour of focused ones (“Should Class 9 Science include the Sāṅkhya tanmātra mapping when teaching the five senses?”).
- Day 9 draft: every student should leave with at least 2 pages and a clear thesis statement.

Grading

- Quiz: standard rubric.
- Paper: see `assessments/rubric.md` — heavy weight on *quality of argument* and *fair representation of counter-arguments*.

If you only have 6 days

- Day 1: Hook + tattva ladder
- Day 2: Tanmātras + Greek comparison
- Day 3: Primary source reading
- Day 4: Modern lens + discussion
- Day 5: Paper drafting
- Day 6: Paper submission + brief quiz

Sources

Sources — Module 2 Panchabhūta (Senior)

At Senior band, ALL citations must be verifiable and properly formatted. Students cite in their paper.

Primary scriptural sources

Citation	Used in	Anchor verse / passage
SB 2.10.3	Day 1, 5	<i>bhūta-mātrendrya-dhiyām janma sarga udāhṛtaḥ ...</i> — defines <i>sarga</i> and enumerates the five gross elements. Direct verse in cache.
SB 3.5.36	Day 2, 3	<i>mahīm gandha-guṇām ādhāt kālamāyāmśa-yogataḥ</i> — earth element acquiring smell as its distinguishing quality.
SB 3.26.36	Day 5 primary source reading	<i>mṛdutvaṁ kaṭhinatvaṁ ca śaityaṁ uṣṇatvaṁ eva ca / etat sparśasya sparśatvaṁ tanmātratvaṁ nabhasvataḥ</i> — tanmātra of touch (air).
SB 3.26.44	Day 3	<i>rasa-mātrād vikurvāṇād ambhaso daiva-coditāt gandha-mātram abhūt</i> — taste transforming into

Citation	Used in	Anchor verse / passage
SB 3.26.50 (paraphrase)	Day 2 quality-count	smell; water giving rise to earth. “Earth has five qualities, water four, fire three, air two, sky one.”

All in vidya-karana RAG cache . rag - cache/. Use the IAST + word-by-word gloss + continuous translation provided.

Secondary scholarly sources

- **Larson, G. J. (1979).** *Classical Sāṅkhya: An Interpretation of its History and Meaning*. Motilal Banarsidass. **The standard reference on Sāṅkhya tattva enumeration.** Use for the 25-tattva ladder on Day 2.
- **Plofker, K. (2009).** *Mathematics in India*. Princeton University Press. **Historical context** of classical Indian intellectual frameworks; useful for situating Sāṅkhya in time.
- **Bryant, E. F. (2009).** *The Yoga Sūtras of Patañjali*. North Point Press. — Modern annotated translation of Patañjali; useful for connecting Sāṅkhya and Yoga.

Comparative sources (Day 4)

- **Empedocles** (c. 490–430 BCE) — Fragments on the four roots: earth, water, fire, air. English translations available in Kirk, Raven & Schofield, *The Presocratic Philosophers* (2nd ed., Cambridge 1983).
- **Aristotle** (c. 4th century BCE) — *On Generation and Corruption*, Book II. Discussion of the four sublunary elements + the celestial *aither* (quintessence).

For both Empedocles and Aristotle, free English translations are available via the Perseus Project (perseus.tufts.edu) and the Internet Classics Archive.

Modern science context

- NCERT Class 11 Chemistry, Chapters 1–3 (atomic structure, classification of elements).
- For plasma and Bose-Einstein condensate as states of matter beyond the classical four: any modern physics survey text.

Citation style for student papers

- For scripture: (SB 3.26.36) inline; full reference in works cited as Bhāgavata Purāṇa, Canto 3, Chapter 26, Verse 36. Translation: A. C. Bhaktivedanta Swami Prabhupada (1972).
- For scholarly works: standard MLA or Chicago. State chosen style on title page.
- For comparative source: (Empedocles, fragment B17) or (Aristotle, GC II.3).
- No fabricated citations. The teacher will check at random.

What to tell students about the source mix

Three different *kinds* of source are used in this module: 1. **Primary scriptural** (verses themselves) — facts about what the tradition says. 2. **Secondary scholarly** (Larson, Plofker,

Bryant) — academic interpretation and historical context. 3. **Comparative** (Empedocles, Aristotle, modern science) — cross-cultural and cross-period comparison.

A strong paper cites all three. A weak paper cites only one type and presents it as the whole picture.

Day 1 — Hook: A Persistent Question

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students articulate what “categorising matter” means and identify three distinct historical frameworks for doing it.

Materials

- Whiteboard
- Periodic table poster
- Printed copy of SB 2.10.3 with IAST + translation (from cache)

Flow

Warm-up (5 min)

Opening provocation on board: “*Categories are not in nature. Categories are imposed on nature. Discuss.*”

Take 3 student responses.

Core (25 min)

1. **Three frameworks, one question.** Write on board:

Framework	Approximate date	Categories
Greek (Empedocles, Aristotle)	5th c. BCE → 4th c. BCE	4 + 1 (quintessence)
Sāṅkhya (India)	likely pre-5th c. BCE, codified c. 4th c. CE	25 tattvas (5 are bhūtas)
Modern periodic table	1869 (Mendeleev) → present	118+ elements

2. **The persistent question:** “*What categories of stuff (and not-stuff) does the world come in?*” Every culture’s natural philosophers tried to answer this. The answers differ because the methods differ and the questions are framed differently.
3. **Today’s anchor verse — SB 2.10.3.** Read in IAST. Then translation: > “*bhūta-mātrendrya-dhiyāṁ janma sarga udāhṛtaḥ ...*” > “The manifestation of the gross elements, the senses, and the intellect is called creation (*sarga*).”

Note the move: the Sāṅkhya verse names the bhūtas (elements), the indriyas (senses), and the dhī (intellect) as products of a single creation event. Modern science talks about atoms; Sāṅkhya talks about a unified scheme that includes the *perceiver*. This is a real structural difference, not just terminology.

Discussion (10 min)

Open question: “*Is the modern periodic table ‘more correct’ than Empedocles’ four elements? More correct in what way?*”

- Let students argue.
- Reinforce: a framework can be more *predictive* (chemistry, atomic structure) without being more *comprehensive* (the Sāṅkhya scheme includes consciousness; modern chemistry doesn’t claim to).

Wrap-up (5 min)

- Preview Day 2: “*Tomorrow we map out the full 25-tattva ladder of Sāṅkhya. The five elements are only the LAST five.*”
- Homework: read the SB 2.10.3 cache entry in full. Note one question you have.

Differentiation

- **Tier up:** Read Kirk, Raven & Schofield’s intro on Empedocles. Be ready to present on Day 4.
- **Tier down:** Stick to the framework-comparison; the primary-source reading deepens on Day 5.

Teacher notes

- Don’t let “modern science is just objectively right” be the resting position. Push: “what does objectively right *mean*? Predictive accuracy? Comprehensiveness? Explanatory power?”
- This module is about *intellectual honesty across frameworks*, not about advocacy for any one of them.

Citation(s) used

- SB 2.10.3 — vidya-karana RAG cache day-01-hook.json#chunks[0]

Day 2 — The Sāṅkhya Tattva Ladder

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students draw and label all 25 tattvas in the Sāṅkhya scheme, and locate the five bhūtas within the full enumeration.

Materials

- Large whiteboard or pinboard
- A4 sheets for students
- Sources excerpt: Larson's *Classical Sāṅkhya* summary (teacher-prepared, 1 page)

Flow

Warm-up (5 min)

- 2 students share their homework question from Day 1.
- Quick recap: "Five elements emerge LAST in Sāṅkhya. Today we see what comes first."

Core (25 min)

1. Build the ladder on the board, slowly:

25. puruṣa (consciousness, the witness)
24. prakṛti (primordial nature, unmanifest)
_____ (evolution begins)
23. buddhi / mahat (intellect)
22. ahaṅkāra (ego, sense of "I")
_____ (branch into three lines)

Mental:	Sense organs:	Action organs:
21. manas (mind) (speech)	20. śrotra (ear)	15. vāk
(hand)	19. tvac (skin)	14. pāṇi
(foot)	18. cakṣus (eye)	13. pāda
(anus)	17. rasanā (tongue)	12. pāyu
(genitals)	16. ghrāṇa (nose)	11. upastha

Tanmātras (subtle elements, 10–6):
10. śabda-tanmātra (sound)
9. sparśa-tanmātra (touch)
8. rūpa-tanmātra (form)
7. rasa-tanmātra (taste)
6. gandha-tanmātra (smell)

Mahābhūtas (gross elements, 5–1):
5. ākāśa (space)
4. vāyu (air)
3. tejas (fire)
2. ap (water)
1. pṛthvī (earth)

(Numbering convention varies in different Sāṅkhya texts; here we count from the most concrete (1 = earth) up to the most abstract (24 = prakṛti) with puruṣa as the silent 25th. Other texts count from puruṣa down.)

2. The key structural insight:

- The five gross elements (*mahābhūtas*) are the LAST and most concrete layer.
- Each is paired with a *tanmātra* (subtle quality, the perceptual essence).
- Each *tanmātra* is what the corresponding sense-organ perceives.
- The whole scheme is internally consistent: the *bhūtas*, *tanmātras*, and *indriyas* line up in a tight one-to-one mapping.

Diagram exercise (10 min)

Students draw the 25-tattva ladder in their notebooks. Use colour to mark the five *bhūtas* distinctly.

Wrap-up (5 min)

- Quick check: “Where do the five elements sit in the ladder?” (At the bottom — the most concrete, most manifest layer.)
- Preview Day 3: “Tomorrow — the *bhūta-tanmātra* pairing in detail, and why the quality-count rule follows.”

Differentiation

- **Tier up:** Read the full first 22 verses of the Sāṅkhya Kārikā in translation. Note one verse you find puzzling.
- **Tier down:** Focus on the *bhūtas* + *tanmātras* layer. The full ladder can be a reference.

Teacher notes

- The 25-tattva count is the *classical Sāṅkhya* count (Īśvarakṛṣṇa’s *Sāṅkhya Kārikā*, c. 4th c. CE). Different versions of Sāṅkhya count differently.
- Be honest about the *non-theistic* character of classical Sāṅkhya. It does NOT include a creator god as a tattva. The Yoga school (*Yoga Sūtras*) adds *Īśvara* as a separate principle; classical Sāṅkhya does not.

Citation(s) used

- SB 3.5.36 — `vidya-karana RAG cache day-02-creation.json#chunks[1]`
- Larson, *Classical Sāṅkhya* (1979), chapters on tattva enumeration.

Day 3 — Quality-Count Rule and the Tanmātras

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students explain the *bhūta–tanmātra* pairing and prove the quality-count rule (1, 2, 3, 4, 5 for sky, air, fire, water, earth) is internally consistent given the emergence sequence.

Materials

- Whiteboard
- Printed SB 3.26.36 + 3.26.44 (IAST + translation, from cache)

Flow

Warm-up (5 min)

- Quick recall: enumeration of the five *bhūtas* with their *tanmātras*.

Core (25 min)

1. The quality-count rule (write boxed):

Each element inherits all the qualities of the elements that emerged before it, AND adds one new quality of its own — the *tanmātra* associated with that element.

Element	Qualities	Source-verse anchor
ākāśa	śabda	SB 3.26 (qualities-of-space)
vāyu	śabda + sparśa	SB 3.26.36 (directly quoted: “softness, hardness, cold, heat — qualities of touch, <i>tanmātra</i> of air”)
tejas	śabda + sparśa + rūpa	—
ap	śabda + sparśa + rūpa + rasa	—
pṛthvī	śabda + sparśa + rūpa + rasa + gandha	SB 3.26.44 (taste → smell as the last new quality)

2. Internal consistency: the scheme is *symmetric and complete*:

- 5 *bhūtas*, 5 *tanmātras*, 5 *indriyas* of perception.
- Each *bhūta* pairs with exactly one *tanmātra*.
- Each *tanmātra* is perceived by exactly one *indriya*.
- The quality count (1, 2, 3, 4, 5) is a *predicted consequence* of the emergence order — not an arbitrary stipulation.

3. Read SB 3.26.36 together (in IAST): > “*mṛdutvaṃ kaṭhinatvaṃ ca śāityaṃ uṣṇatvaṃ eva ca / etat sparśasya sparśatvaṃ tanmātratvaṃ nabhasvataḥ*” > “Softness, hardness, cold, and heat — these are the qualities of touch (*sparśa*), the subtle attribute (*tanmātra*) of air (*nabhasvat*).”

- Word-by-word with class: *mṛdutvaṃ* = softness, *kaṭhinatvaṃ* = hardness, *śāityaṃ* = cold, *uṣṇatvaṃ* = heat. *etat* = these. *sparśasya sparśatvaṃ* = the touch-ness of touch. *tanmātratvaṃ nabhasvataḥ* = the subtle-essence-ness of air.

- Discussion: notice that *sparsā* (touch) here includes thermal sensations (cold, heat). This is broader than modern English “touch” (which usually means pressure/texture). The framework is more inclusive than we’d assume from a casual reading.

Concept check (10 min)

Worksheet question (1 mark each): - Why does *ākāśa* have only one quality? - Why does *prthvī* have five? - What does *sparsā* include that modern English “touch” doesn’t? - Why is the bhūta-tanmātra mapping called “one-to-one”? - If a sixth element existed, how many qualities would it have, and what would they be? (Discussion question.)

Wrap-up (5 min)

- Preview Day 4: “*Tomorrow — Empedocles and Aristotle. Greek and Indian elemental schemes side by side.*”

Differentiation

- **Tier up:** Write a 1-page response to “if a sixth element existed (e.g. ‘information’), where would it sit and what would its tanmātra be?”
- **Tier down:** Focus on the quality-count rule; the bhūta-tanmātra pairing alone is the core skill.

Teacher notes

- The *internal consistency* point matters more than students realise. Many will assume “old framework = primitive.” The Sāṅkhya enumeration is *not* primitive — it’s tightly self-consistent in a way that the Greek four-element scheme isn’t.
- The Empedocles scheme (Day 4) has NO equivalent of the tanmātra-indriya symmetry. That asymmetry is one of the most interesting cross-cultural points in the module.

Citation(s) used

- SB 3.26.36 — `vidya-karana RAG cache day-03-qualities.json#chunks[3]`
- SB 3.26.44 — `vidya-karana RAG cache day-03-qualities.json#chunks[0]`

Day 4 — Empedocles, Aristotle, and the Greek Four

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students compare the Greek (Empedocles + Aristotle) four/five element scheme with Sāṅkhya, identifying at least three structural differences and one structural similarity.

Materials

- Printed extracts:
 - Empedocles fragments on the four roots (Kirk, Raven & Schofield translation)

- Aristotle, *On Generation and Corruption*, II.3 (on the four elements + aither)
- Whiteboard
- Side-by-side comparison worksheet (blank)

Flow

Warm-up (5 min)

- Recall: the 5-bhūta + 5-tanmātra + 5-indriya structure of Sāṅkhya.

Core (25 min)

1. **Empedocles in brief.** Read 1-paragraph paraphrase aloud: > Empedocles (c. 490–430 BCE) proposed four “roots” (*rhizomata*) — earth, water, air, fire. He added two forces — Love (*philia*) that brings them together and Strife (*neikos*) that separates them. The world is the cyclic combination/separation of the four under these forces.
2. **Aristotle in brief:** > Aristotle (4th c. BCE) inherited the four sublunary elements (earth, water, air, fire) but added a *fifth element* — *aither* (the “quintessence”) — which is the substance of the heavenly spheres. The four are characterised by pairs of qualities: hot/cold and wet/dry.

Element	Aristotle’s qualities
Fire	hot + dry
Air	hot + wet
Water	cold + wet
Earth	cold + dry
Aither	none (incorruptible)

3. **Side-by-side comparison.** Build with class on the board:

	Sāṅkhya (India)	Empedocles + Aristotle (Greece)
Nu mb er of ele me nts	5	4 + 1 (quintessence)
Or igi n	Emerge sequentially from <i>prakṛti</i>	Eternal “roots”
W hat dri ve s	Three <i>guṇas</i> (sattva, rajas, tamas)	Love and Strife (Empedocles); the natural motions toward “proper place” (Aristotle)

	Sāṅkhya (India)	Empedocles + Aristotle (Greece)
combination		
Fifth element	<i>Ākāśa</i> — universal medium of space, present everywhere	<i>Aither</i> — celestial substance, only in the heavens
Quantity	1, 2, 3, 4, 5 (predicted from emergence order)	No equivalent — each element gets a pair of qualities (hot/cold + wet/dry)
Rule		
Perceptual mapping (tānmatrā-indriya)	Yes — explicit, symmetric, one-to-one	No equivalent
Inclusion	Yes (puruṣa, 25th tattva)	No
consciousness		

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4. **Discuss the most interesting difference.** Open: “Which of these structural differences feels most significant to you? Why?”

Activity (10 min)

Begin in-class comparison essay draft: - Pick ONE structural difference from the table. - Write a 1-paragraph (5–7 sentence) analysis of why it matters. - This is a draft for the longer Day 8–10 paper.

Wrap-up (5 min)

- Preview Day 5: “Tomorrow — we read SB 3.26.36 closely in IAST, word by word.”

Differentiation

- **Tier up:** Read Empedocles fragment B17 (on Love and Strife) in translation; write 1 paragraph on whether the *guṇa* doctrine in Sāṅkhya is structurally parallel.
- **Tier down:** Focus on Aristotle alone; skip Empedocles for now.

Teacher notes

- The asymmetry on perceptual mapping (Sāṅkhya YES, Greek NO) is the most distinctive cross-cultural point. Push students to notice it.
- Some students will be tempted to conclude “Sāṅkhya is more sophisticated.” Push back: *more sophisticated in what way?* (More internal symmetry, yes. More predictive of chemical behaviour, no.)

Citation(s) used

- Kirk, G. S., Raven, J. E., & Schofield, M. (1983). *The Presocratic Philosophers* (2nd ed.). Cambridge University Press. — Empedocles fragments.
- Aristotle, *On Generation and Corruption*, II.3. (Any standard translation.)

Day 5 — Primary Source Reading + Mid-Module Check

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students read SB 3.26.36 in IAST with word-by-word gloss and submit the formative quiz.

Materials

- Printed handout: SB 3.26.36 in Devanāgarī, IAST, word-by-word gloss, continuous translation

- Formative quiz (quizzes/formative.md Part B)

Flow

Warm-up (5 min)

- Hand out the printed verse. Quick recall: yesterday's Greek comparison.

Reading (15 min)

1. **First read — sound.** Teacher reads the IAST aloud once. Students listen.

*mṛdutvaṃ kaṭhinatvaṃ ca śaityam uṣṇatvaṃ eva ca etat sparśasya sparśatvaṃ
tanmātratvaṃ nabhasvataḥ*

2. **Second read — meaning.** Students follow the word-by-word gloss:

Sanskrit	Gloss
mṛdutvaṃ	softness (nominative neuter)
kaṭhinatvaṃ	hardness
ca	and
śaityam	cold
uṣṇatvaṃ	heat
eva ca	as well
etat	these
sparśasya	of touch (genitive)
sparśatvaṃ	the touch-ness (= the defining quality of touch)
tanmātratvaṃ	the subtle-essence-ness
nabhasvataḥ	of air / wind (genitive; nabhasvat = wind)

Continuous translation: > “Softness, hardness, cold and heat — these are the touch-ness of touch, the subtle essence of air.”

3. **Class discussion (8 min):**

- What surprises you about the categories included under “touch”?
- Note the philosophical move: *sparśasya sparśatvaṃ* (touch-ness of touch) — the verse identifies the defining property of a category by giving its essence a name.
- How is this different from a modern dictionary definition of “touch”?

Formative quiz (15 min)

- quizzes/formative.md Part B.

Wrap-up (5 min)

- Preview Day 6: “*Tomorrow — the modern lens. What modern science adds, and what it explicitly drops.*”

Differentiation

- **Tier up:** Write a 1-paragraph response to “what is gained when philosophy gives an *essence* a name?”
- **Tier down:** Focus on the meaning. The IAST pronunciation is bonus.

Teacher notes

- Don’t require pronunciation perfection. The win is *engagement with the meaning at primary-source level*.
- This is many students’ first time reading a Sanskrit verse. Honour the moment.

Citation(s) used

- SB 3.26.36 — `vidya-karana RAG cache day-03-qualities.json#chunks[3]`

Day 6 — The Modern Lens

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students identify three things modern science *adds* to the picture that Sāṅkhya does not have, and three things Sāṅkhya *includes* that modern science explicitly excludes.

Materials

- Periodic table poster
- NCERT Class 11 Chemistry, Chapter 1 (atomic structure) — printed extract
- Whiteboard

Flow

Warm-up (5 min)

- Hand back yesterday’s quiz. Brief review of common errors.

Core (25 min)

1. **What modern science adds:**
 - **Atomic structure** — protons, neutrons, electrons. The periodic table’s 118 elements are atom-based, not state-of-matter-based.
 - **Predictive chemistry** — combinations follow rules (valence, electronegativity) we can calculate.

- **Subatomic and field-level descriptions** — quarks, electromagnetic field, weak and strong forces. None of this is in Sāṅkhya.
 - **Quantitative measurement** — mass, density, melting point. Sāṅkhya is qualitative.
2. **What Sāṅkhya includes that modern science explicitly excludes:**
- **Empty space (*ākāśa*) as a category** — modern physics treats vacuum as the absence of matter, not as a distinct category.
 - **A perceptual mapping** — *tanmātra-indriya* one-to-one symmetry. Modern science treats perception as biology (neuroscience), not as a metaphysical layer.
 - **Consciousness in the scheme** — *puruṣa* (the witness) is part of the 25-tattva enumeration. Modern science either bracket consciousness (default in chemistry) or treats it as emergent from matter (in some neuroscience interpretations).
3. **The honest framing:** these are different KINDS of inquiry, not different *qualities* of inquiry.

Inquiry	Modern science	Sāṅkhya
“What predicts chemical behaviour?”	Strong	Weak
“What categories does the everyday world come in?”	Decent (states of matter)	Strong
“How does perception fit with reality?”	Modern: neuroscience; not in chemistry	Strong (<i>tanmātra-indriya</i>)
“Where does consciousness fit?”	Mostly bracketed	Central (<i>puruṣa</i>)

Reflection writing (10 min)

5-minute quick-write: “For one question above where Sāṅkhya is ‘Strong’ but modern science is ‘Mostly bracketed’ or ‘Weak,’ do you think this is a real intellectual asset of the Sāṅkhya scheme, or a leftover of pre-scientific guessing? Defend your position in 4–5 sentences.”

Discussion (5 min)

3 students share their position. Note the diversity of views.

Wrap-up (5 min)

- Preview Day 7: “Tomorrow — group debate. Should IKS be integrated into mainstream science class?”

Differentiation

- **Tier up:** Read Thomas Nagel’s “What Is It Like to Be a Bat?” — write 1 paragraph on whether Sāṅkhya’s *tanmātra-indriya* symmetry maps onto Nagel’s “subjective experience” question.

- **Tier down:** Stick to the comparison table; the quick-write can be 2–3 sentences.

Teacher notes

- Push students past “old vs new” framing into “different scopes, different methods.”
- Some students may genuinely believe consciousness is “just neurons” — that’s a defensible position. Some may believe it’s irreducible. Both are taken seriously in contemporary philosophy of mind; the module shouldn’t decide for them.

Citation(s) used

- NCERT Class 11 Chemistry, Chapter 1.
- (Optional) Nagel, T. (1974). “What Is It Like to Be a Bat?” *Philosophical Review*, 83(4), 435–450.

Day 7 — Discussion: Should IKS Be in Science Class?

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students articulate and defend a position on whether classical Indian intellectual frameworks should be taught in mainstream science class, with at least one citation and one steel-manned counter-argument.

Materials

- Whiteboard
- Stopwatch
- Score sheet for the oral assessment (rubric below)

Flow

Warm-up (3 min)

- Brief: today is structured debate. Three positions on the table. Pick one. Defend it. Acknowledge the other two fairly.

The three positions (5 min — clarify, don’t argue yet)

Write on board:

Position	One-line claim
A — Integration	IKS frameworks should be taught alongside modern science in mainstream science class.
B — Separation	IKS should be taught in cultural studies or philosophy, not in science class.
C — Hybrid	IKS should be referenced briefly in science class as historical context, but not taught as parallel content.

Debate (25 min)

- Divide class into 3 groups by position (let students self-select; balance roughly).
- 5 min: each group prepares — strongest 2 arguments, 1 steel-manned response to the other two positions.
- 12 min: structured debate. Each group has 4 min.
 - 2 min: main argument.
 - 1 min: response to the other 2 positions.
 - 1 min: closing.
- 8 min: open floor cross-questions.

Oral assessment (during the debate)

Teacher uses this rubric to score 5 students chosen at random:

Criterion	4	3	2	1
Position clarity	Crystal clear with 1-line summary	Clear	Mostly clear	Unclear
Use of evidence	Cites at least 1 specific source	Cites generally	Vague references	No evidence
Counter-argument	Steel-mans an opposing view	Acknowledges another view	Mentions another view	Ignores
Listening	Responds to specific points raised	Responds generally	Sticks to script	Doesn't engage

Wrap-up (10 min)

- Take 3 student reflections: “*Did anyone’s mind change during the debate?*”
- Note: the goal is not to “win” but to *understand the strongest version of each position*.
- Preview Day 8: “*Tomorrow you start your paper. Use this debate to sharpen your own thesis.*”

Differentiation

- **Tier up:** Take a position you DISAGREE with and defend it. (The hardest form of argument.)
- **Tier down:** Pre-prepare your 2-minute argument the night before.

Teacher notes

- This is where some students discover they don't really know what they think. That's the point.
- If the debate gets heated, slow it down: "*Steel-man the other side. What's their best argument?*"
- The Day 7 oral grade flows into the project rubric — not as a separate score.

Citation(s) used

- All sources from `sources.md` are fair game during the debate.

Day 8 — Project Session 1: Outline Your Paper

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students choose a focused paper topic, draft a one-paragraph thesis, and produce a working outline.

Materials

- Project brief (`assessments/project-brief.md`)
- Rubric (`assessments/rubric.md`)
- Laptop or notebook for drafting

Flow

Warm-up (5 min)

- Hand out the rubric. Read aloud the *thesis* and *counter-argument* criteria.

Core: outlining (30 min)

Sprint 1 (10 min) — Topic. Each student writes: - Working title (1 line) - One-paragraph thesis (3–4 sentences) - 3 sources they'll cite (must include one primary scriptural, one secondary scholarly, one comparative or modern)

Teacher rotates: 60 sec per student, single question: "*What's the counter-argument someone could make against your thesis?*"

Sprint 2 (10 min) — Outline. Each student writes section headings + 1-line sub-points: - Introduction (with thesis statement) - Section 1 (anchor claim + source) - Section 2 (anchor claim + source) - Counter-argument and response - Conclusion

Sprint 3 (10 min) — Begin drafting. Each student starts the introduction or one body paragraph.

Wrap-up (10 min)

- Each student exchanges outline with one buddy.
- 3-min mutual feedback: 1 strength + 1 fix.
- Outline + thesis submitted at the door.

Homework

- Draft 1–2 more paragraphs at home. Bring everything tomorrow.

Differentiation

- **Tier up:** Add a 4th source — a contemporary academic article (search Google Scholar).
- **Tier down:** Topic can be narrow (“a comparison of one Sāṅkhya tanmātra concept with the modern five-senses model”). Narrower is fine.

Teacher notes

- Common pitfall: too-broad topic. Push for specificity early.
- A weak thesis sounds like “Indian science is interesting.” A strong thesis sounds like “The Sāṅkhya tanmātra–indriya symmetry offers a model that 19th-century European psychophysics independently rediscovered.”

Day 9 — Project Session 2: Draft + Counter-Arguments

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Each student reaches at least 2 drafted pages including a fully-articulated counter-argument and response.

Materials

- Drafts in progress
- Rubric

Flow

Warm-up (5 min)

- Brief: today’s focus is the counter-argument. The rubric rewards a *fair* presentation of the strongest opposing view.

Core (30 min)

Sprint 1 (15 min) — Draft body. Quiet writing. Aim for 1.5–2 pages drafted.

Sprint 2 (15 min) — Counter-argument workshop. - Pair with a buddy. - Read each other’s drafts so far. - Each pair argues: “*Take my thesis. Give me the strongest counter-argument you*

can think of.” - The student takes notes on the counter-argument and decides how to respond in their paper.

Wrap-up (10 min)

- 3 students share their best-discovered counter-argument.
- Outline-to-date submitted at door (revised since Day 8).

Homework

- Finish a complete draft at home tonight. Bring tomorrow.

Differentiation

- **Tier up:** Construct TWO counter-arguments — one philosophical, one practical.
- **Tier down:** Submit a clear outline + 1 paragraph if a full draft is too much.

Teacher notes

- Some students will skip the counter-argument because it feels like “weakening” their position. Reframe: a strong paper anticipates and answers objections.

Day 10 — Summative Quiz + Paper Submission

Module: Pañcabhūta (Senior) · **Time:** 45 min (paper accepted up to end of day)

Learning objective

Students complete the summative quiz and submit their 5-page argumentative paper.

Materials

- Summative quiz (quizzes/summative.md)
- Stopwatch

Flow

Quiz (35 min)

- Distribute. Closed-book.
- 35 marks. Multiple choice, short answer, primary-source reading question, comparison question, open response.

Paper submission (10 min)

- Hand in printed papers (or upload digitally).
- Teacher quick-checks each for:
 - Title page with name + thesis
 - Page count (4.5–5.5 pages target)
 - Works-cited page with at least 3 sources

- Any paper missing a piece is flagged for late penalty.

Wrap-up

- Brief class closure.
- Reflection slip submitted (3 questions from `assessments/project-brief.md`).

Homework

- None. Module complete. (Paper marked within 1 week.)

Differentiation

- **Tier up:** Students who submit early can begin reading for the next module's prerequisites.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark papers in two passes: first pass for clarity and structure; second pass for source quality and counter-argument fairness.
- Return marked papers within 5 school days with detailed comments.

Citation(s) used

- All sources verified.

Module 2 (Senior) — Formative Quizzes

Part A — Day 3 (8 min, 5 questions)

1. (2 marks) Write the 5 bhūtas in emergence order with their tanmātras.
2. (1 mark) What does *sparśa* include in SB 3.26.36 that modern English “touch” usually does NOT?
3. (1 mark) Why does *pṛthvī* have 5 qualities while *ākāśa* has only 1?
4. (1 mark) The Sāṅkhya scheme has how many total tattvas?
5. (1 mark) Where do the five bhūtas sit in the tattva ladder — at the top, middle, or bottom?

Answer key

Q	Answer	Marks
1	ākāśa-śabda, vāyu-sparśa, tejas-rūpa, aprasa, pṛthvī-gandha	2 (1 if order correct, 1 if pairings correct)
2	Cold and heat (thermal sensations)	1
3	Earth emerges last, inheriting all prior qualities AND adding its own (gandha). Space emerges first, has no prior	1

Q	Answer	Marks
	qualities to inherit, adds only śabda.	
4	25	1
5	Bottom (most concrete, most manifest layer)	1

Total Part A: 6 marks.

Part B — Day 5 (20 min, 6 questions including primary source)

1. (3 marks) Read this verse (already provided in handout):

mṛdutvaṃ kaṭhinaṭvaṃ ca śaityam uṣṇatvaṃ eva ca / etat sparśasya sparśatvaṃ tanmātratvaṃ nabhasvataḥ

Identify (1 mark each): a) Which bhūta is named? _____ b) Which tanmātra is identified as belonging to it? _____ c) What four specific qualities are listed as examples? _____

2. (2 marks) Translate *sparśasya sparśatvaṃ* word-by-word. What philosophical move does this phrase make?

3. (2 marks) Compare: Aristotle's *aither* and Sāṅkhya's *ākāśa*. Name ONE crucial difference.

4. (2 marks) Why is the bhūta–tanmātra–indriya mapping called “one-to-one” and “symmetric”?

5. (2 marks) State one thing modern science ADDS that Sāṅkhya does not have, and one thing Sāṅkhya INCLUDES that modern science excludes.

6. (3 marks) Open: in 3–4 sentences, articulate your current best position on whether Sāṅkhya's tanmātra-indriya mapping is “philosophically interesting” or “empirically obsolete.” Defend.

Answer key

Q	Answer	Marks
1a	vāyu (air; nabhasvat)	1
1b	sparśa (touch)	1
1c	softness, hardness, cold, heat (mṛdutvaṃ, kaṭhinaṭvaṃ, śaityam, uṣṇatvaṃ)	1
2	“touch-ness of touch” — the defining quality of touch. The move: naming the essence of a category so it can be philosophically tracked.	2
3	Aether: ONLY in heavens, incorruptible, celestial. Ākāśa:	2

Q	Answer	Marks
	EVERYWHERE, the universal space-medium, terrestrial AND celestial.	
4	5 bhūtas, 5 tanmātras, 5 indriyas — each in one-to-one pairing. No bhūta has two tanmātras; no tanmātra has two perceivers.	2
5	Examples: modern adds atomic structure / predictive chemistry; Sāṅkhya includes empty space (ākāśa) as a category / consciousness (puruṣa) in the scheme	2
6	Personal answer. Marked: 1 for clear position, 1 for ≥1 specific reason, 1 for clarity.	3

Total Part B: 14 marks.

Module 2 (Senior) — Summative Quiz

Time: 35 min · **Marks:** 35 · **Closed-book**

Name + class at top. Answer all questions. Section E requires a structured argument.

Name: _____ **Class:** _____

Section A — Multiple choice ($5 \times 1 = 5$ marks)

- Classical Sāṅkhya enumerates how many tattvas? a) 5 b) 8 c) 25 d) 108
- The tanmātra of *tejas* (fire) is: a) śabda b) sparśa c) rūpa d) gandha
- Empedocles's four "roots" are: a) earth, water, fire, air b) earth, water, fire, ether c) sattva, rajas, tamas, ākāśa d) hot, cold, wet, dry
- Aristotle's "quintessence" (aither) is: a) the substance of perception b) the substance of the celestial spheres c) the same as Sāṅkhya's ākāśa d) the fifth state of matter
- Which of the following is in the Sāṅkhya scheme but NOT in the modern periodic table? a) Hydrogen b) Water c) Empty space (ākāśa) d) Fire

Section B — Sanskrit reading (4 marks)

- Given the verse: *mṛdutvaṃ kaṭhīnatvaṃ ca śāityam uṣṇatvaṃ eva ca / etat sparśasya sparśatvaṃ tanmātratvaṃ nabhasvataḥ*

- a) (1 mark) Which bhūta does *nabhasvat* refer to?
- b) (1 mark) Translate *sparsāsya sparsātvam* word-by-word.
- c) (2 marks) What does this verse say about the relationship between qualities (mṛdutvam etc.), tanmātras, and bhūtas?
-

Section C — Short answer (3 × 3 = 9 marks)

7. (3 marks) Why does the quality-count rule (1, 2, 3, 4, 5 for sky, air, fire, water, earth) follow *necessarily* from the emergence sequence?
8. (3 marks) Name three structural differences between the Sāṅkhya scheme and the Greek (Empedocles + Aristotle) scheme.
9. (3 marks) What does modern science predict that Sāṅkhya cannot? Give one specific example.
-

Section D — Comparison (5 marks)

10. (5 marks) Fill in the comparison table:

Property	Sāṅkhya	Modern periodic table
# categories	_____	_____
Includes empty space?	_____	_____
Treats consciousness?	_____	_____
Method	_____	_____
Best for question	_____	_____

Section E — Argumentative response (12 marks)

11. In 1–2 pages, argue your position on the following question:

“Modern science has rendered the Sāṅkhya five-element framework obsolete except as cultural history. Discuss.”

Your answer should: - State a clear position (agree, disagree, partially). - Cite at least 2 sources (primary scriptural, secondary scholarly, comparative, or modern science). - Anticipate and respond to one counter-argument. - Avoid the false dichotomy (“either Sāṅkhya is correct OR science is correct”).

Use the back of this sheet and an additional sheet if needed.

Answer key

Q	Answer	Marks
1	c) 25	1
2	c) rūpa	1
3	a) earth, water, fire, air	1
4	b) the substance of the celestial spheres	1
5	c) Empty space (ākāśa)	1
6a	vāyu (air)	1
6b	“touch-ness of touch” — sparśasya = of touch (genitive); sparśatvam = the touch-ness, the defining quality of touch	1
6c	The qualities (softness, etc.) ARE the tanmātra (subtle essence) of the bhūta (air). The verse identifies a category’s essence with the perceptual quality associated with it.	2
7	Each bhūta inherits the qualities of all bhūtas that emerged before it (because they manifest into the new bhūta) AND adds its own tanmātra. So the count = emergence position.	3
8	Examples: Sāṅkhya is 5+5+5 symmetric, Greek isn’t; Sāṅkhya has ākāśa, Greek’s aither is celestial-only; Sāṅkhya emerges from prakṛti, Greek roots are eternal; Sāṅkhya has tanmātra-indriya mapping, Greek doesn’t. (Any 3.)	3
9	Specific examples acceptable: predicts that combining 2 hydrogen + 1 oxygen makes water; predicts NaCl melting point; etc.	3
10	5 vs 118; Yes vs No; Yes (puruṣa) vs No; Sensory observation vs Experimental measurement; Everyday matter and space vs Predictive chemistry	5
11	Marked on: clear position (3), source citations (3), counter-argument fairly stated and addressed (3), avoidance of false dichotomy (3).	12

Total: 35 marks.

Grading bands (paper combined → out of 100, with paper at 65): - 85–100: Distinction - 70–84: Merit - 55–69: Pass - 0–54: Resit

Homework

Homework — Module 2 Pañcabhūta (Senior)

Day 1 → Day 2

Reading: SB 2.10.3 from the cache (IAST + translation). Write 1 question you have about the verse.

Time: 20 min.

Day 2 → Day 3

Reading: Larson, *Classical Sāṃkhya*, opening chapter on tattva enumeration (teacher's 1-page summary).

Time: 25 min.

Day 3 → Day 4

Reading: Empedocles fragment B17 (translation in handout). Note: how does Love-and-Strife differ from the Sāṃkhya guṇa doctrine?

Time: 25 min.

Day 4 → Day 5

Practise: Try reading SB 3.26.36 in IAST aloud (in handout). Don't worry about perfect pronunciation.

Time: 15 min.

Day 5 → Day 6

Writing: Submit the in-class quiz follow-up — 1 paragraph reflection on what reading a Sanskrit verse felt like.

Time: 20 min.

Day 6 → Day 7

Preparation for debate: Choose your position (Integration / Separation / Hybrid). Prepare your 2-minute opening argument + 1 steel-manned response.

Time: 30 min.

Day 7 → Day 8

Submit project plan: working title, thesis paragraph, 3 source citations.

Time: 30 min.

Day 8 → Day 9

Draft: Aim for 1.5–2 pages drafted at home.

Time: 60–90 min.

Day 9 → Day 10

Finalise: Complete draft, proofread, format. Submit tomorrow.

Time: 60–90 min.

Writing Prompts

Writing Prompts — Module 2 Pañcabhūta (Senior)

Eight prompts at essay-level depth.

Reflection prompts

- 1. (Day 1)** Categories are conventions imposed by minds, not features of nature. Defend or refute in 1 page, using examples from the periodic table and the Sāṅkhya tattva enumeration.
- 2. (Day 6)** Modern science excludes consciousness from chemistry. Sāṅkhya includes puruṣa as the 25th tattva. Is this an asset or a liability of the Sāṅkhya scheme? 1 page.

Explanatory prompts

- 3. (Day 3 or Day 5)** In 1 page, explain to a college-level reader why the quality-count rule (1, 2, 3, 4, 5) is a *predicted consequence* of the Sāṅkhya emergence sequence, not an arbitrary stipulation. Use SB 3.26.36 as a primary source.
- 4. (Day 4)** Compare and contrast the role of Empedocles's Love-and-Strife with the role of the three guṇas (sattva, rajas, tamas) in Sāṅkhya. 1 page.

Argumentative prompts

- 5. (Day 7 or Day 8 — could be paper topic)** Argue for or against the claim: “Sāṅkhya's bhūta-tanmātra-indriya symmetry is structurally more complete than Aristotle's element-quality scheme.” 2 pages, with sources.

6. (Senior research) Argue: “The Sāṅkhya scheme’s inclusion of ākāśa as a positive category (rather than as absence-of-matter) anticipates a question that modern field theory addresses but did not solve until the 20th century.” 2 pages.

Cross-disciplinary prompts

7. (Day 6) Pick a contemporary debate (consciousness studies, panpsychism, the hard problem of consciousness) and write 1 page on how Sāṅkhya’s tattva enumeration speaks to it — or fails to.

8. (Stretch) Write a 2-page comparison of the *guṇa* doctrine in Sāṅkhya with the *element* doctrine in classical Chinese philosophy (wǔxíng — Wood, Fire, Earth, Metal, Water). Use one specific source from each tradition.

How to grade essays

- Strong: clear thesis, specific citations, fair counter-argument, original thinking, clean prose.
- Adequate: clear thesis, some citations, weak counter-argument, mostly competent prose.
- Weak: vague thesis, no citations, no counter-argument, structural issues.

Activity 01 — Frameworks Comparison Table

Module: Pañcabhūta (Senior) · **Day:** 4 · **Time:** 25 min within Day 4’s 45

Goal

Students fill in a structured comparison table contrasting Sāṅkhya, Greek (Empedocles + Aristotle), and modern frameworks across 7 dimensions.

Materials

- Printed comparison worksheet (one per student)
- Reading extracts on hand

Instructions

Working alone first, then with a buddy, fill in the table comparing the three frameworks.

For each cell, write a one-line answer. Where the framework has no equivalent, write “—” and we’ll discuss why in the next session.

Table (student-facing)

Dimension	Sāṅkhya	Greek (Empedocles + Aristotle)	Modern science
Number of elements			
Origin			

Dimension	Sāṅkhya	Greek (Empedocles + Aristotle)	Modern science
(eternal? emerge? created?)			
What drives combination			
Includes empty space?			
Includes consciousness?			
Perceptual mapping?			
Method (observation , experiment, philosophy) ?			

Discussion (last 10 min)

After 15 min of filling in: pair with a buddy and compare. Where do you disagree? Why?

Teacher asks: which row revealed the most interesting cross-cultural difference?

Differentiation

- **Tier up:** Add an 8th row of your own choosing.
- **Tier down:** Focus on the first 4 rows; the others can be discussed orally.

Clean-up

- Tables collected — used in the Day 8 paper outline.

Project Brief

Project Brief — Module 2 Pañcabhūta (Senior)

Type: 5-page argumentative paper · **Marks:** 65 · **Due:** End of Day 10

The question

Choose ONE of the following or propose your own (with teacher approval):

1. **Should classical Indian intellectual frameworks be taught alongside modern science in mainstream science class?**
2. **Is the Sāṅkhya tanmātra-indriya symmetry a philosophical achievement that modern psychophysics later rediscovered, or an unrelated coincidence?**
3. **Does the Sāṅkhya inclusion of ākāśa as a positive category (not “absence of matter”) anticipate any question that modern field theory addresses?**
4. **Compare the role of guṇas (sattva, rajas, tamas) in Sāṅkhya with the role of Love-and-Strife in Empedocles. What does each scheme explain that the other doesn’t?**

Required elements

Your paper must include:

1. **Clear thesis statement** in the opening paragraph.
2. **At least one primary scriptural source** quoted in IAST + translation (from the RAG cache files in .rag-cache/).
3. **At least one secondary scholarly source** (Larson, Plofker, Bryant, or another academic source you locate).
4. **At least one cross-cultural or modern comparative source** (Empedocles, Aristotle, modern science).
5. **A steel-manned counter-argument** and your considered response to it.
6. **Works-cited page** with proper formatting (MLA or Chicago).

Format

- 5 pages (4.5–5.5 acceptable)
- Double-spaced, 12pt
- Title page with name, class, thesis statement
- IAST diacritics used correctly

Schedule

- Day 7: choose topic.
- Day 8: outline + thesis submitted.
- Day 9: draft (≥2 pages) shared with buddy.
- Day 10: final paper submitted.

Rubric

See `rubric.md`. Familiarise yourself BEFORE Day 8.

Self-reflection (1 paragraph, submitted with paper)

1. What did you set out to argue, and what did you actually argue?
2. What is the strongest counter-argument you encountered, and how did you respond?
3. If you had three more days, what would you research deeper?

Rubric

Paper Rubric — Module 2 Pañcabhūta (Senior)

Paper marks: 65 (out of 100 with summative quiz 35).

Criteria

Criterion	4 — Distinction	3 — Merit	2 — Pass	1 — Resit	Weight
Thesis clarity	Crystal-clear, specific, defensible thesis stated in opening	Clear thesis	Thesis present but vague	No clear thesis	× 3
Use of primary scripture	At least 1 verse quoted in IAST + translation; correctly interpreted	Cites a verse; interpretation adequate	Mentions scripture vaguely	No primary scripture	× 2
Use of secondary scholarship	Cites at least 1 academic source (Larson, Plofker, Bryant); used substantively	Cites a scholarly source	Vague academic reference	None	× 2
Cross-cultural / modern comparison	Strong comparison with at least 1 specific point	Comparison present	Surface-level comparison	None	× 2
Counter-argument	Steel-manned opposing view + reasoned response	Counter-argument present	Acknowledged opposing view	Ignored	× 2
Avoidance of false dichotomy	Refuses “either-or”; holds multiple frameworks	Mostly nuanced	Some both-and reasoning	Reduces to either-or	× 1
Originality	Original synthesis; surprising insight	Independent thinking visible	Mostly synthesises others	Largely paraphrased	× 1
Prose / structure	Polished, well-paragraphed, clean citations	Clear, mostly clean	Some clarity issues	Hard to follow	× 1

Max raw: 56. Paper score = (sum × weights) × 65/56.

Format requirements (binary, deduct only)

- 5 pages ± 1 (deduct 5 marks if outside)

- Title page + thesis (deduct 5 if missing)
- Works-cited with proper format (deduct 5 if missing or formatting wrong)
- IAST diacritics used correctly where Sanskrit appears (deduct 3 if absent)

Late submission

- -5 marks per school day, capped at -20.

Plagiarism

- A paper that closely matches an online source without citation goes to 0 on Originality AND on the source criteria.
- Single-day resubmission allowed once.